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High-Dimensional Representations in Data Science

The Curse and Blessing of Dimensionality

Introduction to High-Dimensional Data Science

Data Science involves modeling phenomena as numerical data, understanding the geometric and statistical properties of this data, and developing tools to analyze it.

What is Data Science?

Data Science encompasses:

- Modeling phenomena as quantitative numerical data
- Understanding the geometric and statistical properties of data
- Analyzing the space in which this data exists
- Developing analytical tools with an understanding of their underlying assumptions

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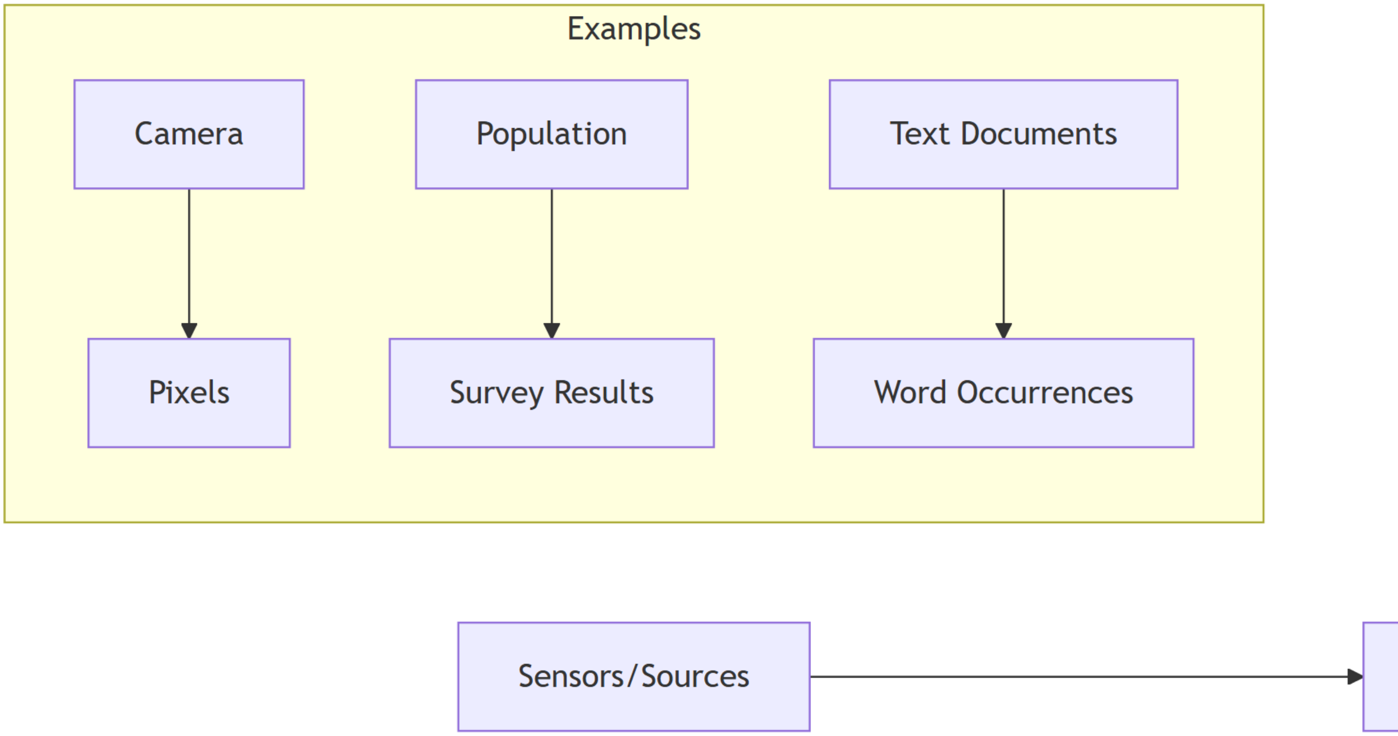
Relationship with Machine Learning

Data Science and Machine Learning are closely related:

- **Data Science**: study of representation spaces
- **Machine Learning**: operations within these representation spaces

Both fields are complementary and often synonymous in practice.

Typical Data Flow



Properties of Representation Spaces

The representation space Ω is typically a subset of   where D is the dimension.

This space has several important properties:

- It's a vector space that can be equipped with an inner product
- The inner product · , · induces a norm || · ||
- The norm induces a distance d(· , ·)
- (Ω, d) forms a metric space where learning can occur

Fundamental Questions:

- What happens when D increases?
- What impact does this have on data modeling?
- How does this affect learning?

These questions lead to the concept of the **curse of dimensionality**.

The Curse of Dimensionality

Understanding the Phenomenon

In high dimensions, our geometric intuition based on 2D and 3D becomes misleading. Counter-intuitive phenomena emerge.

Empty Sphere and "Spiky" Cube

The High-Dimensional Sphere Paradox

Consider a hypersphere B_D(r) of radius r centered at the origin, inscribed in a hypercube C_D(r) = [-r, r] .

The volume of the hypersphere is given by:

$$\mathrm{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$$

The volume of the hypercube is:

$$\mathrm{vol}(C_D(r)) = (2r)^D$$

The sphere-to-cube volume ratio decreases rapidly with D:

$$\text{Ratio} = \frac{\mathrm{vol}(B_D(r))}{\mathrm{vol}(C_D(r))} = \frac{\pi^{D/2}}{D 2^{D-1} \Gamma(D/2)}$$

The surprising result:

$$\lim_{D\to\infty} \frac{\mathrm{vol}(B_D(r))}{\mathrm{vol}(C_D(r))} = 0$$

Intuitive Interpretation

Imagine sampling points uniformly at random in the cube:

- In 2D, about 78.5% of points fall within the inscribed circle
- In 3D, this percentage drops to about 52.4%
- In 10D, it's only 0.025%
- In 100D, it's infinitesimal

Practical consequence: If you sample uniformly in a hypercube, the inscribed sphere will be essentially empty. Volume concentrates in the "corners" of the cube.

Volume Distribution: The "Gaussian Egg"

Gaussian Concentration in High Dimensions

For a normal distribution N(, Σ) in D dimensions, consider the density along a radius r.

If coordinates are i.i.d. random variables X ~ N(0,1), then:

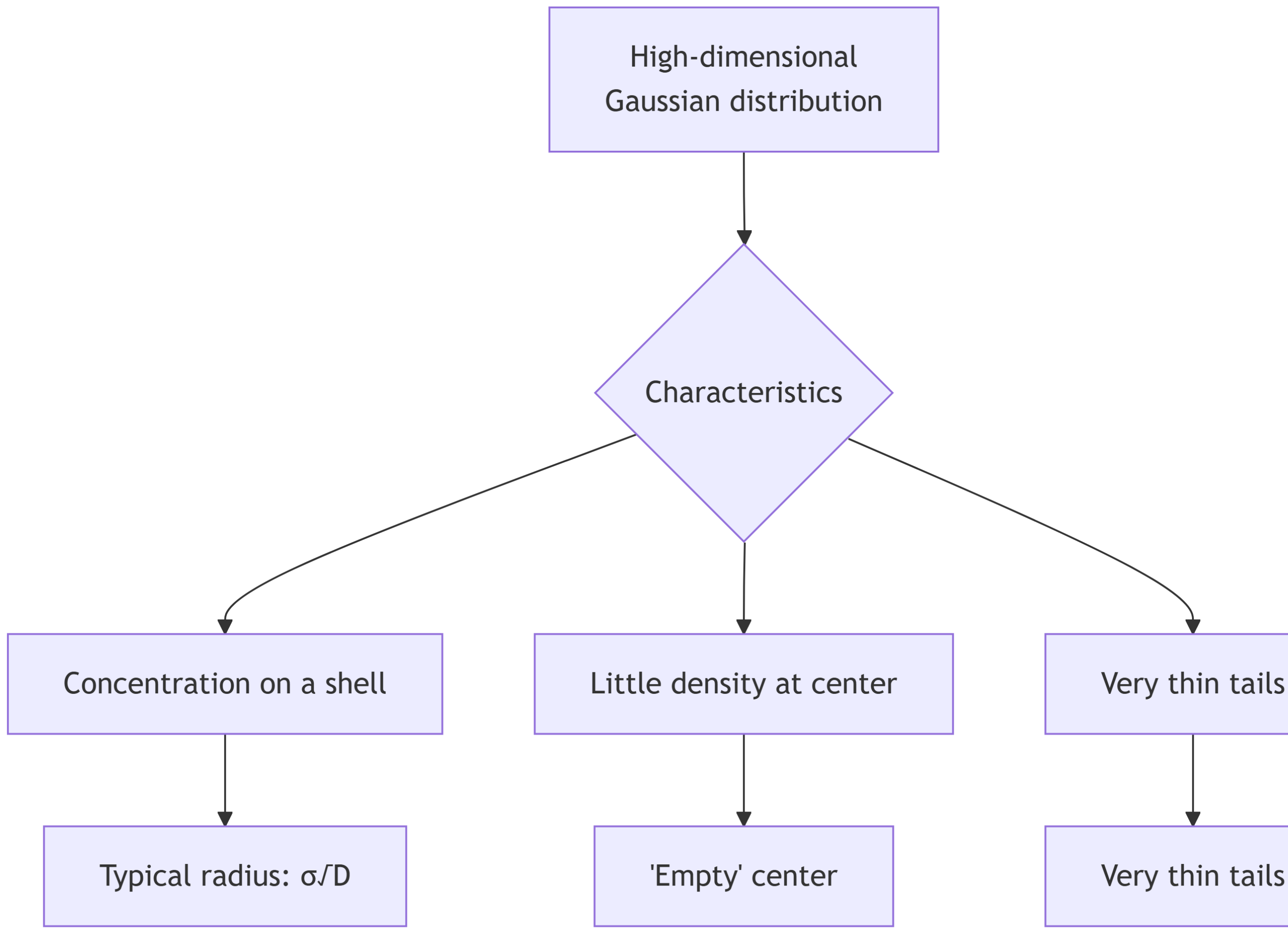
$$R^2 = \|X\|_2^2 = \sum_{i=1}^{D} X_i^2 \sim \chi^2(D)$$

The expectation of R^2 is:

$$\mathbb{E}[R^2] = D$$

Important implication: Most of the probability mass concentrates on a spherical shell of radius $\sqrt{D}$.

Visualizing the Phenomenon



Distance Concentration

Beyer et al.'s Theorem

Given a dataset X Ω, a point x and its k nearest neighbors V_X (x). Let D_min and D_max be the distances to the nearest and farthest neighbors.

Theorem: If $\lim_{D\to\infty} \mathrm{Var}\left(\frac{d(x,y)^p}{\mathbb{E}(d(x,y)^p)}\right) = 0$ for 0 < p < ∞, then for any > 0:

$$\lim_{D\to\infty} P\left[\frac{D_{\max} - D_{\min}}{D_{\min}} \leq \varepsilon\right] = 1$$

Practical Implications

This result has significant consequences:

1. **Loss of discriminative power:** All neighbors appear at roughly the same distance
2. **Meaningless neighborhood structures:** k-NN and -NN structures lose their utility
3. **Algorithm difficulties:** Distance-based discrimination becomes nearly impossible

Concrete example: In classification, if all points are equidistant, distinguishing different classes by proximity becomes very difficult.

Quasi-Orthogonality

Angle Concentration

In high dimensions, random vectors tend to be nearly orthogonal.

For three points X, Y, Z drawn independently, the angle between (X-Y) and (Z-Y) satisfies:

$$\mathbb{E}\cos(\theta) = \frac{1}{2}$$

This implies /3 (60°).

Interpretation: Any random triangle in high dimension is nearly equilateral.

Impact on the Unit Sphere

For the unit sphere B_D, if X is a random point on its surface:

- If Y is sampled on the sphere: $\mathbb{E}\cos(\theta) = \frac{1}{2}$ /3
- If Y is sampled in the ball: $\mathbb{E}\cos(\theta) = 0$ /2

Geometric consequence: The volume of the sphere is concentrated at its equator, regardless of which direction is considered "north".

Hubness: The Hub Phenomenon

Definition and Mechanism

In high dimensions, some points (called "hubs") become the nearest neighbors of a disproportionate number of other points.

Why this occurs:

- Distance concentration makes relative distances less discriminative
- Some points happen to be at "average" distances from many other points
- These points thus frequently appear in nearest neighbor lists

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Negative Implications

The hubness phenomenon causes several problems:

1. **Algorithm bias:** Hubs dominate neighborhood search results
2. **Classification problems:** Hubs may appear as representatives of multiple classes
3. **Interpretation difficulty:** Neighborhood relationships become hard to interpret

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Concentration Inequalities

Fundamental Theoretical Tools

Concentration inequalities quantify how measures concentrate around their mean in high dimensions.

Markov's Inequality

For a non-negative random variable X:

$$P(X > \epsilon) \leq \frac{\mathbb{E}X}{\epsilon}$$

Intuition: The smaller the expectation, the less likely X is to exceed a large threshold.

Chebyshev's Inequality

For a random variable with mean and variance ^{2}:

$$P(|X - \mu| > \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

Application: In high dimensions, distances often have decreasing variance, explaining their concentration.

Hoeffding's Inequality

For independent variables X [a, b]:

$$P(|\bar{X} - \mu| > \epsilon) \leq 2e^{-2D\epsilon^2/(b-a)^2}$$

where $\bar{X} = \frac{1}{D}\sum_{i=1}^{D} X_i$.

Importance: This inequality explains the rapid convergence of averages in high dimensions.

The Blessing of Dimensionality

Positive Aspects of High Dimensions

Counterintuitively, high dimensions also offer advantages that can be exploited.

Laws of Large Numbers

Weak Law of Large Numbers

For a set of D i.i.d. variables {X } [D] with E[X] = , for any > 0:

$$\lim_{D \to \infty} P(|\overline{X} - \mu| < \varepsilon) = 1$$

where $\overline{X} = \frac{1}{D} \sum_{l=1}^{D} X_l$.

Important Implications

1. **Guaranteed convergence:** Estimators converge to their true value
2. **Foundation for estimation:** Solid theoretical basis for statistical estimation
3. **Stability:** Results become more stable with increasing dimension

Central Limit Theorem: The rate of convergence is given by:

$$Z_D = \frac{\sqrt{D}}{\sigma}(\overline{X} - \mu) \sim N(0, 1)$$

if Var(X) = ^{2}.

Random Projections

Johnson-Lindenstrauss Lemma

Statement: Given X Ω, there exists a linear map p: → such that for any 0 < < 1 and for all i, j [N]:

$$(1-\varepsilon)\|x_i - x_j\| \leq \|p(x_i) - p(x_j)\| \leq (1+\varepsilon)\|x_i - x_j\|$$

with:

$$d = O\left(\frac{\log N}{\varepsilon^2}\right)$$

Crucial point: The reduced dimension d does not depend on D!

Practical Applications

1. **Efficient dimensionality reduction:** Reduce dimension without losing too much information
2. **Approximate nearest neighbor search:** Locality-sensitive hashing (LSH) methods
3. **Compression:** More efficient storage and processing of data

Key advantage: By exploiting the quasi-orthogonality of vectors in high dimensions, we can create random bases that preserve distances.

Importance in Data Science

Impact on Practical Methods

Understanding high-dimensional phenomena is crucial for designing effective algorithms.

Sampling

Main Challenges

- **Data sparsity:** Exponential number of samples needed to cover the space
- **Density estimation:** Concentration inequalities guide estimator precision
- **High rejection rate:** In methods like rejection sampling, the rejection rate approaches 100%

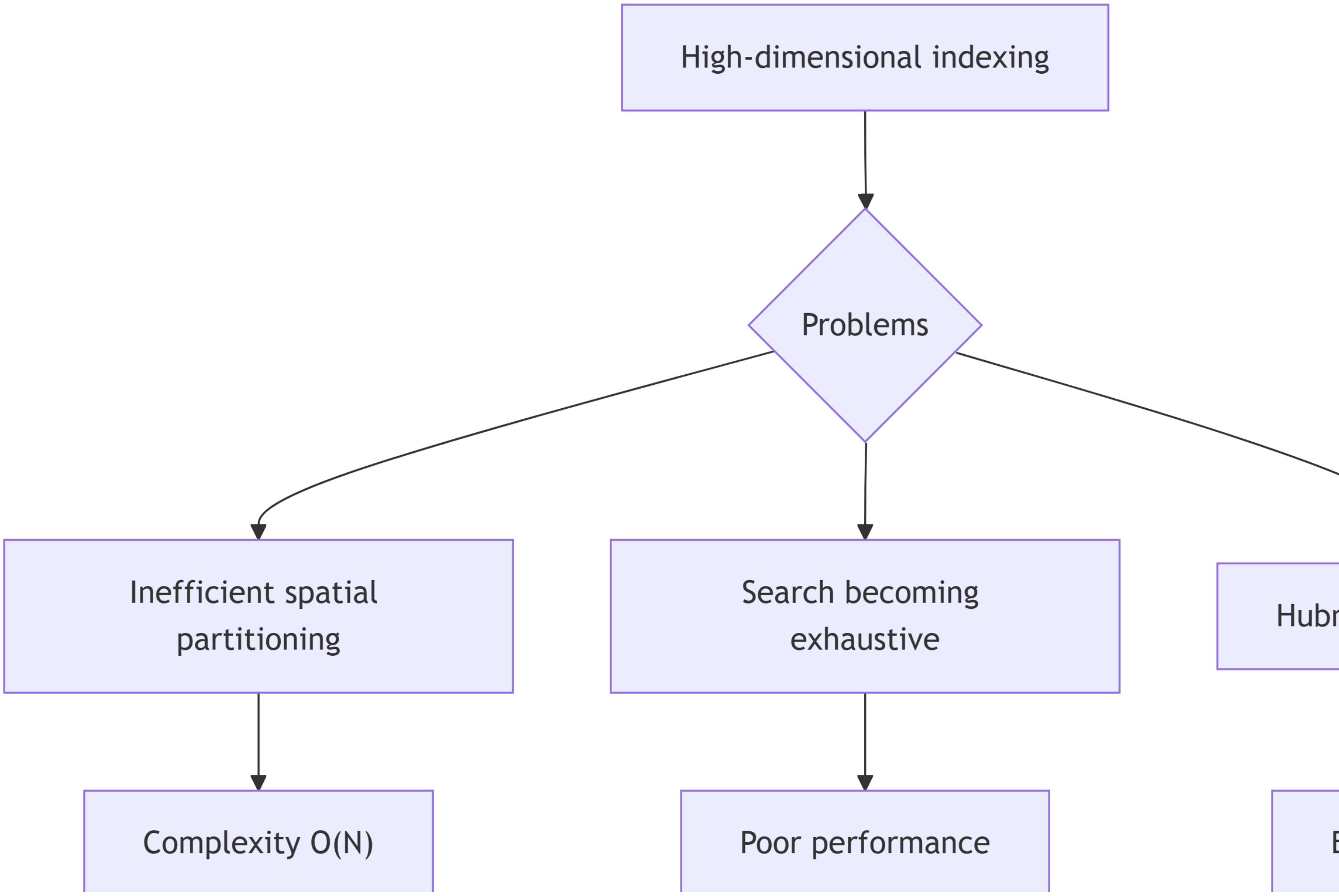
Consequence: Naive sampling methods become impractical in high dimensions.

Indexing and Search

Problems with Spatial Structures

Spatial partitioning structures (like kd-trees) face difficulties:

1. **Structure disappearance:** If all neighbors are equidistant, partitioning loses its meaning
2. **Cells too large:** To cover the space, cells must reach the size of the bounding sphere
3. **Exhaustive search:** Complexity becomes O(N) instead of O(log N)



Should We Abandon Data Science in High Dimensions?

Why Methods Still Work

Contrary to expectations, data science methods work in practice despite the curse of dimensionality:

1. **Real data is not uniform:** It occupies lower-dimensional subspaces
2. **Features are correlated:** The independence assumption is rarely true
3. **Underlying structure:** Data often lies on low-dimensional manifolds

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Strategies to Overcome the Curse

- **Dimensionality reduction:** PCA, autoencoders, t-SNE
- **Feature selection:** Identify relevant dimensions
- **Metric learning:** Learn a distance adapted to the data
- **Specific methods:** Designed for high dimensions (like isolation forests)

Synthesis and Conclusions

Key Points to Remember

1. **Geometry changes radically:** Our 2D/3D intuition doesn't apply
2. **Data becomes sparse:** Exponential samples needed
3. **Distances concentrate:** All points appear equidistant
4. **Quasi-orthogonality emerges:** Random directions are nearly orthogonal
5. **Hubness phenomenon appears:** Some points dominate neighborhood searches

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Balanced Assessment

Curse:

- Difficulties with sampling, indexing, search
- Loss of discriminative power
- Need for enormous amounts of data

Blessing:

- Rapid convergence of estimators
- Possibility of efficient random projections
- Improved statistical stability

Further Questions

1. **Conceptual:**
   - What is the intrinsic dimension of a dataset?
   - How does the curse of dimensionality affect clustering methods?
2. **Technical:**
   - How does the Johnson-Lindenstrauss lemma help circumvent the curse?
   - Which methods are robust to high dimensions?
3. **Practical:**
   - How to detect if your data suffers from the curse of dimensionality?
   - What strategies to adopt for very high-dimensional data?

Future Perspectives

Research continues on:

- Non-linear dimensionality reduction methods
- Algorithms specialized for high dimensions
- Learning on low-dimensional manifolds
- Probabilistic approaches robust to dimension

Final lesson: High dimensionality is not a fatality, but it requires a deep understanding of the phenomena at play and appropriate methods.